2014 9647 H2 Chemistry

Paper 1 Answer Key:

1	B	11	C	21	A	31	B
2	C	12	C	22	D	32	B
3	A	13	B	23	D	33	B
4	A	14	B	24	D	34	B
5	A	15	B	25	D	35	D
6	B	16	D	26	C	36	A
7	A	17	C	27	D	37	A
8	C	18	A	28	C	38	B
9	C	19	C	29	B	39	C
10	B	20	B	30	B	40	A

Paper 1 Worked Solutions

<table>
<tr>
<td>Q1 Answer: B
^{9}B proton number is 4
Nucleon number is 9
Neutrons present = 9-4 = 5

Amt of Be present in 0.09 g = (0.09)/9 = 0.01 mol
Amt of neutrons present = 0.01 x 5 = 0.05 mol
Number of neutrons present = 0.05 L</td>
<td>Q2 Answer: C
The mole ratio of ClO_2 and OH^- must be the same.
Students should be familiar with the following oxidation state of chlorine in +1 state (ClO^-) and +5 state (ClO_3^-).

Chlorine typically do not exist in +2 state. While +7 (ClO_4^-) may appear unfamiliar to the students

Working within the scope of Group VII
Using a chemical equation to deduce the answer:
$2ClO_2 + 2OH^- \rightarrow H_2O + ClO_2^- + ClO_3^-$

Option C will give a logical answer.</td>
</tr>
<tr>
<td>Q3 Answer: A
Students need to convert the skeleton structure into molecular formula.

$C_6H_{12}O_3 + 15/2\ O_2 \rightarrow 6CO_2 + 6H_2O$</td>
<td>Q4 Answer: A
The intermolecular forces of attraction in water and alcohol is hydrogen bond.

For methoxymethane, it’s shape is bent which is unsymmetrical and will result a net dipole moment which is not zero thus resulting in permanent dipole-permanent dipole interaction.</td>
</tr>
<tr>
<td>Q5 Answer: A
The correct respond for the following options is as such
<table>
<tr><td>Option</td><td>No if e^- pairs around central atom (bond pair & lone pair)</td><td>No. of lone pair around central atoms</td><td>Shape</td></tr>
<tr><td>A</td><td>2</td><td>2</td><td>Bent (non linear)</td></tr>
<tr><td>B</td><td>2</td><td>2</td><td>Bent not tetrahedral</td></tr>
<tr><td>C</td><td>3</td><td>1</td><td>Bent not trigonal planar</td></tr>
<tr><td>D</td><td>3</td><td>1</td><td>Bent not linear</td></tr>
</table>
</td>
<td>Q6 Answer: B
Statement A is only applicable to ideal gas
Statement C is only applicable to real gas (intermolecular forces of attraction cannot be ignored)
Statement D is cannot be used for both cases.</td>
</tr>
</table>

Q7 Answer: A

$$\Delta H_{latt} \alpha \frac{q^+ q^-}{r^+ + r^-}$$

ionic radius of Cs^+ >> ionic radius of Na^+
ionic radius of Cl^- >> ionic radius of F^-

value of lattice energy of CsCl should be smallest value.

Q8 Answer: C

$\Delta H_{rxn} = -1273 – [6(-394) + 6(-286)]$
$= -1273 – [-2364 -1716]$
$= +2807$ kJ mol^{1}

Comparing the reactants and products
Reactants contains gas and liquid whereas products consist of formation of a solid and gas.
The formation of the solid will result in an decrease in entropy resulting in negative ΔS

Q9 Answer: C

As stated by the question the above procedure must be an oxidation process

$(SCN)_2 + 2e \rightleftharpoons 2SCN^-$ *x V* (E_{oxd})

From *Data Booklet*

$Cl_2 + 2e \rightleftharpoons 2Cl^-$ +1.36 V (E_{red})

$Br_2 + 2e \rightleftharpoons 2Br^-$ +1.07 V (E_{red})

$I_2 + 2e \rightleftharpoons 2I^-$ +0.54 V (E_{red})

In order of reaction to happen with Cl_2 and Br_2, using
$E_{cell} = E_{red} – E_{oxd}$
E_{cell} value must be positive.

If x = -1.27, all will react
If x = -0.77, all will react
If x = +0.77, reaction with Cl_2 and Br_2 will result in positive E_{cell} but not for the case of I_2

If x = +1.27 reaction with Cl_2 only will result in positive E_{cell} but not for the case of Br_2 and I_2

Q10 Answer: B

Rate = $k[P][Q]^2$

For 1st experiment: Rate = $k[P][Q]^2$

For 2nd experiment: Rate = $k[2P][Q/2]^2$
$= ½ k[P][Q]^2$

Rate will be 2 times faster thus the volume of gas for 2nd experiment is 50 cm^3

Q11 Answer: C

To determine the order of reaction with respect to hydrogen ions, the $[H^+]$ should be varied for various experiments while keeping the other reagents constant. Therefore C is the correct answers.
A and D are incorrect as both measure the change in concentration of H^+ over time. This is not viable as H^+ is a catalyst and will be regenerated. Hence it concentration should not change over time.

Q12 Answer: C

$n_{H^+} = \frac{10}{1000} \times 0.01 = 0.0001$ mol

$[H^+] = \frac{0.0001}{\frac{100}{1000}} = 0.001$

pH = -lg 0.001 = 3

Q13 Answer: B

	$2H_2$ (g) +	CO (g) $\rightleftharpoons$	CH_3OH (g)
I / mol	2.0	1.0	0
C/ mol	-x	-x/2	+ x/2
E/ mol	2.0-x	1.0- x/2	x/2

Concentration of CO at equilibrium = $\frac{1-\frac{x}{2}}{0.5}$ $moldm^{-3}$

Q14 Answer: B

Refer to Gp II lecture notes. Flame colour of barium is green.

Q15 Answer: B

A: Z is A*l*
B: Z is P
C: Z is A*l* or P
D: Z is A*l*
B does not fit with the other 3

Q16 Answer: D

Refer to transition metal lecture notes.
Blue: $[Cu(H_2O)_6]^{2+}$
Yellow: $[CuCl_4]^{2-}$

Q17 Answer: C

Refer to Gp VII lecture notes. Reaction of Cl_2 with hot aq NaOH results in the disproportionation of chlorine.

Q18 Answer: A

Down group,
A: charge density of metal cation ↓. A is correct.
B: reactivity of element with water ↑. B is incorrect.
C: Solubility of oxide in water ↑. C is incorrect.
D: Thermal stability of nitrate ↑. D is incorrect.

Q19 Answer: C

Organic compound is

Dibasic → contains 2 –COOH groups

Non-cyclic

Contains no alkene groups.

Relative M_r of 146

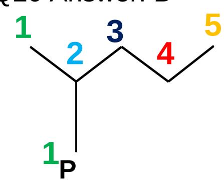
M_r of $(-CH_2)_n$ = Total M_r – 2x (M_r –COOH)

146 – 2(12+16+16+1) = 56

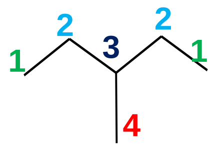
n = 56/(12+2) = 4

Total no of carbons = carbons from $(-CH_2)_n$ + carbons from (-COOH) = 4 + 2 = 6

Q20 Answer: B



P



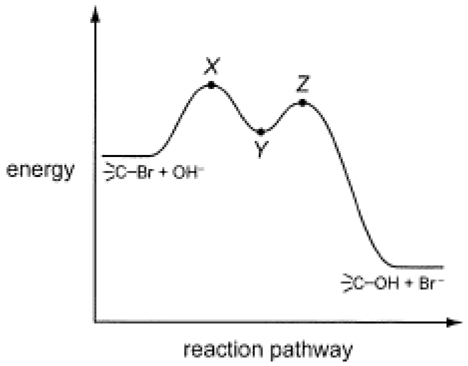
Q

Types of hydrogen = no of possible mono-substituted products

Types of hydrogen in P = 5

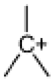
Types of hydrogen in Q = 4

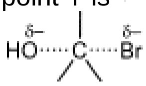
Q21 Answer: A

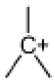


The reaction pathway indicates an S_N1 reaction which is two steps.

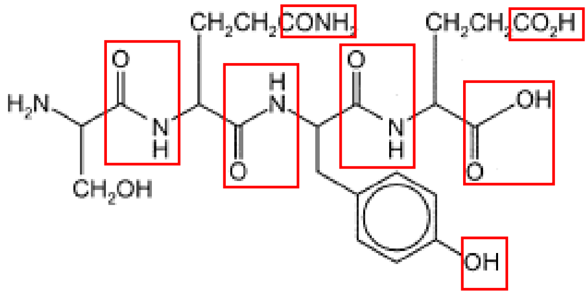
Statement A is correct as it indicates the Br group is leaving.

Statement B is wrong as point Y is  not X

Statement C is wrong as  is only applicable for S_N2 reaction

Statement D is wrong as point Y is  not Z

Q22 Answer: D

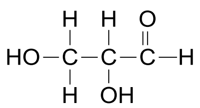


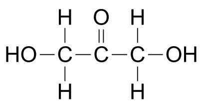
There are 7 sites which can react with NaOH under reflux.

The first 4 boxes from the left will undergo basic hydrolysis.

The 3 boxes from the right will undergo neutralisation.

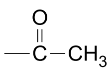
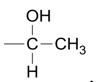
Q23 Answer: D

2,3-dihydroxypropanal  $C_3H_6O_3$

1,3-dihydroxypropanone  $C_3H_6O_3$

Statement A is wrong as only 2,3-dihydroxypropanal contains chiral carbon.

Statement B is wrong as only 2,3-dihydroxypropanaln gives the silver ppt with Tollens' reagent.

Statement C is wrong as both don't contain  or .

Statement D is correct as both share the same formula.

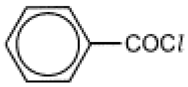
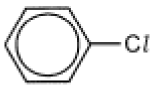
Q24 Answer: D

$$CH_3OH \xrightarrow[\text{step 1}]{\text{HBr, heat}} CH_3Br \xrightarrow[\text{step 2}]{(CH_3)_3P} (CH_3)_4P^+Br^-$$

Step 1 involves the nucleophilic substitution of –OH with Br

Step 2 should be nucleophilic substitution rather than electrophilic substitution (involves benzene). The $(CH_3)_3P$ is functioning as a nucleophile. The product is a quaternary salt.

Q25 Answer: D

Cpd	r.t.p with $AgNO_3$	Reflux with NaOH then $AgNO_3$
 (Acyl chloride)	White ppt	White ppt
 (Halogenoarene)	No ppt	No ppt
CH_3COCl (Acyl chloride)	White ppt	White ppt
$CH_3CH_2CH_2CH_2Cl$ (Halogenoalkane)	No ppt	White ppt

The chlorine in halogenoalkanes will only be substituted out with NaOH (aq), heat. The subsequent addition of $AgNO_3$ will result in the formation of white ppt of AgCl.

Acyl chloride hydrolyses in water to form Cl^- with form white ppt of AgCl when $AgNO_3$ is added.

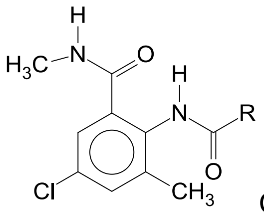
Halogenoarenes are resistant to nucleophilic substitution as the p-orbital of the Cl atom overlap with delocalised electron cloud of the benzene ring.

Q26 Answer: C

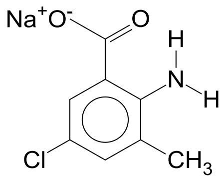
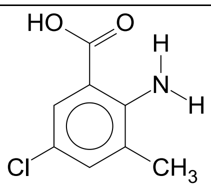
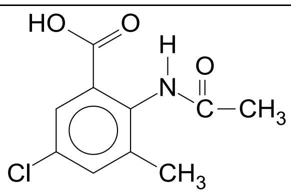
ala-gly
lys-ser
ser-gly-ala
met-ala
gly-lys

Possible structure:
met-ala-gly-lys-ser-gly-ala

Q27 Answer: D



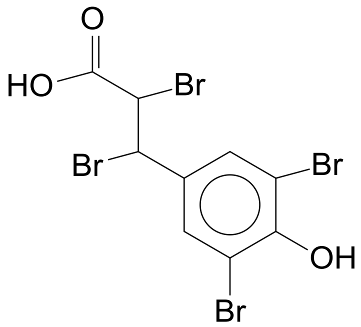
Compound J

J heated with NaOH(aq), Basis hydrolysis will occur, if there are present of carboxylic acid or phenol, neutralisation will also happen	
Solution neutralised and K is isolated	 Compound K
K warmed with CH_3COCl to produce L	

Q28 Answer: C

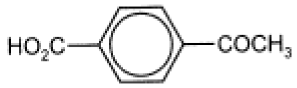
A	$C_6H_5CH_2CHO$ + Fehling (Oxidation)	Blue solution decolorised, reddish brown ppt seen
B	$C_6H_5CHCH_2$ + cold $KMnO_4$ (oxidation)	Purple $KMnO_4$ decolorised.
C	C_6H_5OH + NaOH (aq) Neutralisation	No colour change
D	Phenol + dilute nitric acid	White ppt will be observed.

Q29 Answer: B



Maximum number of bromine atoms is 4.

Q30 Answer: B



1. With Fehling's solution	No observation made as the compound do not undergo oxidation with Fehling's solution.
2. With 2,4-DNPH	Orange ppt is seen.

Thus, only 2 is correct.

Q31 Answer: B
S: $[Ne]3s^2 3p^4$
No. of unpaired electrons = 2

Ti: $[Ar]3d^2 4s^2$
No. of unpaired electrons = 2
Ni: $[Ar]3d^8 4s^2$
No. of unpaired electrons = 2
Co: $[Ar]3d^7 4s^2$
No. of unpaired electrons = 1

Q32 Answer: B
1 is true.
$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$

2 is true
NH_3: no of electrons = 7 + 3 = 10
NH_4^+: no of electrons = 7 + 4 – 1 = 10

3 is incorrect.
NH_4^+ does not have a lone pair of electrons to form dative bonds with metal ion.

Q33 Answer: B
In general, the solubility of Group II sulfates decreases down the group, as the extent of the decrease in the magnitude of L.E. outweighs that of the decrease in the magnitude of ΔH_{hyd} of M^{2+}(g).

Underlying concept:
$\Delta H_{sol} = \sum \Delta H_{hyd} - \text{L.E.}$

Option 3 (ionisation energies) does not affect ΔH_{sol}.

Q34 Answer: B
1 is incorrect.
Standard condition: **298 K**, 1 atm

2 is incorrect.
$H_2SO_4 \equiv 2H^+$
1 mol dm^{-3} $H_2SO_4 \equiv$ **2 mol dm^{-3} H^+** (not standard condition)

Q35 Answer: D
1 is true. Metal chlorides are ionic chlorides.

2 is not true. Sc^{3+}: [Ar], a very large amount of energy is needed to disrupt the noble gas configuration, Sc^{4+} and Sc^{5+} will not form readily.

3 is not true. Sc^{3+}: [Ar] has zero d-electrons. No d-d transition may occur.

Q36 Answer: A
X is arsenic, Group V.

Hence, 1 and 2 are correct.
3 is correct. Arsenic oxide is considered a non-metal oxide, hence acidic oxide. FYI, arsenic oxide is amphoteric.

Q37 Answer: A
Option 1: nucleophilic substitution of RX
Option 2: nucleophilic substitution of RX
Option 3: Elimination of HBr from RX

All reagents and conditions are also accurate.

Q38 Answer: B
1 is correct. Unburnt hydrocarbons are oxidised to less harmful gases such as CO_2 and H_2O in a catalytic converter.

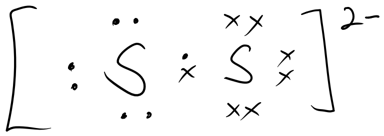
2 is correct. NO is reduced to N_2 by CO in a catalytic converter.

3 is incorrect. CO is not a product of a catalytic converter.

Q39 Answer: C
Hydrogen bonding occurs in the secondary, tertiary and quaternary structures of protein.

Q40 Answer: A
1 and 3 are correct. E and F molecular formulae are $C_{12}H_{10}N_2O_2$; G molecular formula is C_6H_5NO.

Paper 2 Worked Solutions

2(a)	**(i)**	Mg^{2+} and SO_4^{2-}	**(ii)**	Fe^{2+}, Mn^{2+} and Cu^{2+} ions
2(b)	$[H^+]$ in sea water = $10^{-7.8}$ = 1.58×10^{-8} mol dm^{-3} $[H^+]$ in hydrothermal vent water = $10^{-4.3}$ = 5.01×10^{-5} mol dm^{-3}			
2(c)	$SO_4^{2-} + 10H^+ + 8e \rightarrow H_2S + 4H_2O$			
2(d)	**(i)**	Oxidised: sulphur Reduced: hydrogen	**(ii)**	
	(iii)	Transition metal complexes are coloured due to the presence of incompletely filled d-orbitals in the metal ion. In a complex, the presence of **<u>disulphide ligands</u>**, split the 3d orbital of Fe^{2+} into two energy level. This is called d-orbital splitting. A **<u>d electron absorbs certain wavelengths of light energy (or absorption of the photon)</u>** from the **<u>visible region of the electromagnetic spectrum</u>**. The **<u>d electron</u>** undergoes **<u>d-d transition</u>** and is **<u>promoted to a higher energy d orbital</u>**. The **<u>remaining wavelengths</u>** are **<u>transmitted</u>** and appear as the **<u>colour observed (golden yellow ppt).</u>**		
2(e)	**(i)**	Relative atomic mass of an element is the weighted average mass of an atom in the naturally occurring isotopic mixture, compared to $^1/_{12}$ **the mass of an atom of carbon-12 isotope**. OR Relative atomic mass is the ratio of the **<u>average mass of an atom</u>** of an element to **<u>1/12 of the mass of an atom of carbon-12</u>**.		
	(ii)	Let the percentage of ^3He in the mixture be $x\%$ $$\frac{x}{100}(3.0160293) + \frac{100-x}{100}(4.0026033) = 4.0025959$$ Solving $x = 7.50 \times 10^{-4}$ Percentage of ^3He in mixture = 7.50×10^{-4} %		

3(a)

pV=nRT

$$\frac{n}{V} = \frac{p}{RT} = \frac{250 \times 10^3}{(8.31)(25+273)}$$

$\frac{n}{V} = 101\ mol\ m^{-3}$ = 0.101 mol dm^{-3}

3(b) (i)

Use equation in reaction 2:

$K_c = \frac{[H_2CO_3]}{[CO_2]}$ *(Note: water should not be included in the K_c expression)*

1.70 x 10^{-3} = $\frac{[H_2CO_3]}{0.084}$

$[H_2CO_3]$ = 1.43 x 10^{-4} mol dm^{-3}

(ii)

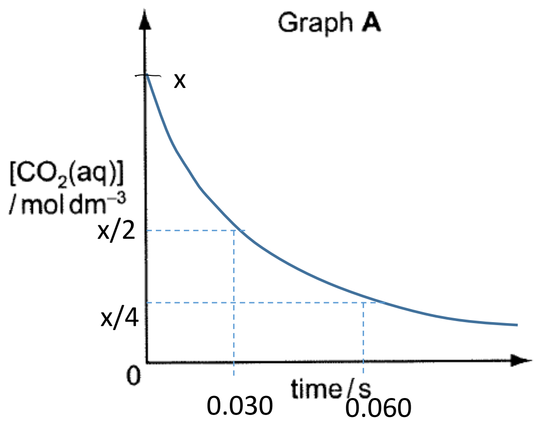
When the bottle is opened, **<u>CO_2 gas escapes</u>**.

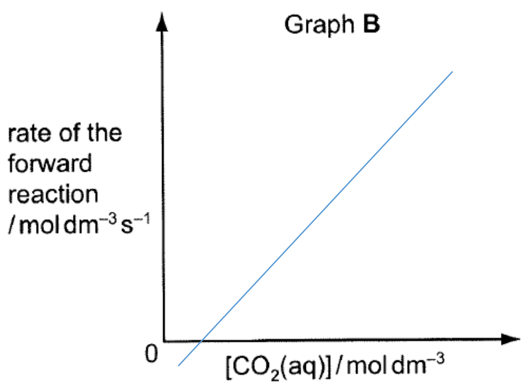
By Le Chatelier's Principle, **<u>position of equilibrium in reaction 1 shift to the left to compensate the decrease in amount of CO_2 gas</u>**.

As such, concentration of **<u>CO_2 decreases causing the equilibrium position in reaction 2 to shift left</u>**. This in turn causes **<u>equilibrium of H_2CO_3 (aq) to decrease which will result in the position of equilibrium in reaction 3 to shift left.</u>**

This result in the [H$^+$] to decrease.
pH= -lg [H$^+$], pH of the drink increases.

3(c) (i)





(ii)

$t_{1/2}$ = ln2 /k
0.030 = ln 2 / k
k = 23.1 s^{-1}

(iii)

1-0.99 = $\left(\frac{1}{2}\right)^{\frac{time\ taken}{0.030}}$

$$\frac{time\ taken}{0.030} = \frac{ln0.01}{ln0.5}$$

time taken = 0.2s

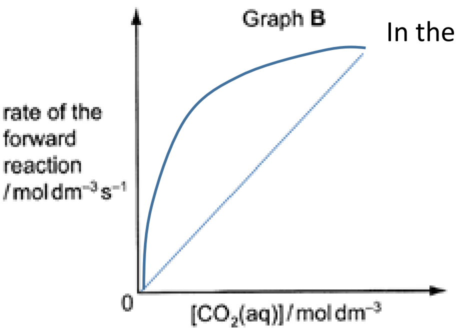
[Note: use the formula $ln\left(\frac{N_t}{N_o}\right) = \frac{t}{t_{1/2}} ln\frac{1}{2}$

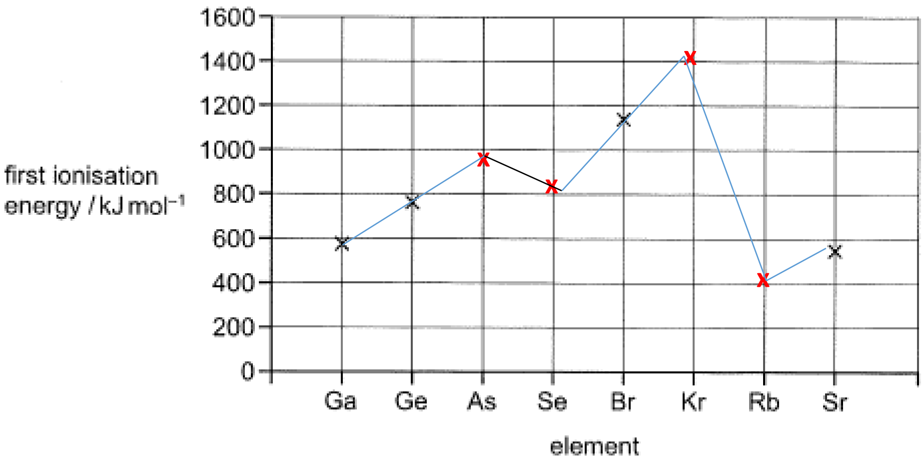
where:
N_o is the initial amount of substance
N_t is the amount of substance remaining
t is time
$t_{1/2}$ is half life

Alternative approach :

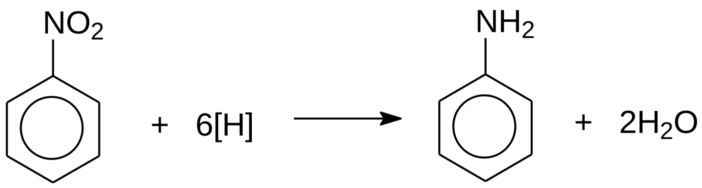
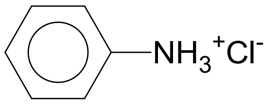
Proportion of carbonic acid formed = 1 – (1/2)^n
Where n is the number of half life

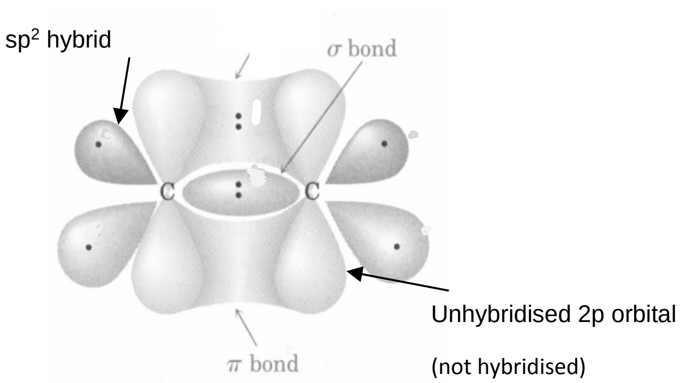
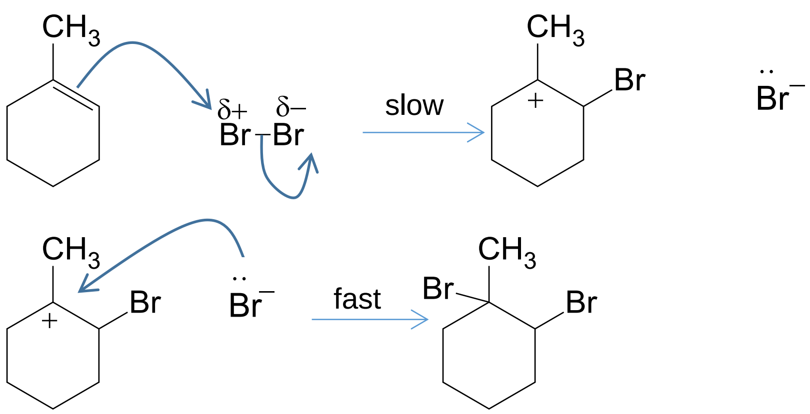
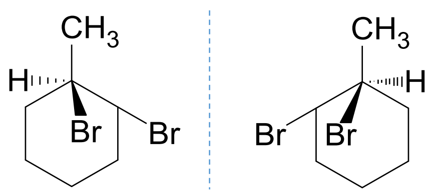
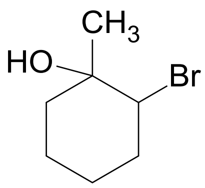
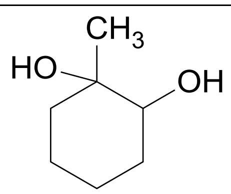
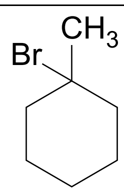
0.99 = 1 – (1/2)^n
n = ln(0.01) ÷ ln(0.5)
n = 6.643
Time taken = 6.643 x 0.03 = 0.199 s

3(d)	**(i)**	 In the presence of enzyme
	(ii)	Effect on [substrate] on rate of reaction For a fixed amount of enzyme, there is a finite number of active sites. At low [CO_2] (substrate), the sites are not filled and rate = k[CO_2] This implies that the rate is directly proportional to [CO_2] and reaction is deem first order w.r.t CO_2 As [CO_2] increases, the increase in rate becomes smaller. Lastly, increase in [CO_2] has no effect on the rate of reaction and is said to be zero order with respect to CO_2. This is because the enzyme is saturated with its substrate and each enzyme binding sites has a substrate attached to it. All the enzyme molecules are continuously catalysing the conversion of the substrate to the product.

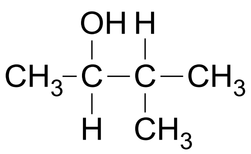
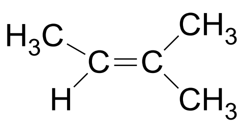
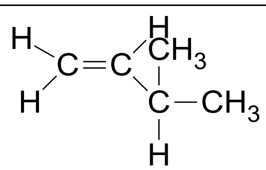
4(a)	
4(b)	Across the period , first ionisation energy generally increases which explain the rise from **Ge to As** and **Br to Kr**. Reason: - Across the period, **nuclear charge increase** - Shielding effect is **similar** since successive elements in the period have an additional electron in the same valence shell. - Effective nuclear charge increases. - **More** energy is required to overcome the **stronger** electrostatic forces of attraction between the nucleus and the valence electron to be removed. The first IE of Se is lower than As. Reason: - There is **interelectronic repulsion** between the pair of electrons in the **doubly-filled 4p orbital** of Se.

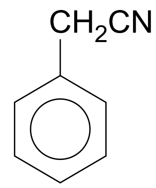
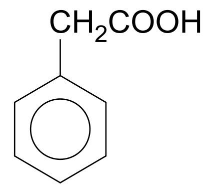
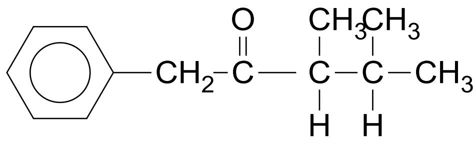
	- **Less** energy is required to overcome the **weaker** electrostatic forces of attraction between the nucleus and the **paired** valence **4p** electron in Se compared to the **unpaired** valence **4p** electron in As. The first IE of Rb is lower than Kr Reason: - It involves the removal of a **5s electron for Rb** compared to a **4p electron for Kr**. - **Less energy** is required to remove the 5s electron as compared to a 4p electron as the **5s electron is further** from the nucleus and thus experiencing weaker electrostatic forces of attraction.

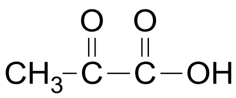
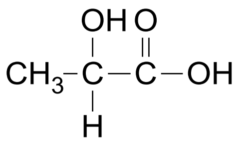
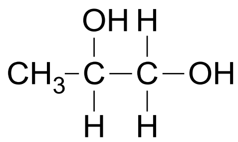
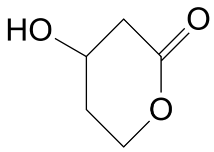
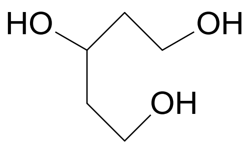
5(a)	
5(b)	Product is a salt  after reduction in acidic medium. Excess concentrated NaOH is added to produce phenylamine which can be separated from the mixture via distillation.
5(c)	Amount of nitrobenzene = 4.84/123.0 = 0.03935 mol Nitrobenzene is the limiting reagent. Nitrobenzene ≡ phenylamine Theoretical mass of phenylamine = 0.03935 x 93 = 3.66 g Percentage yield = (Actual yield/ Theoretical yield) x 100% = (2.64/3.66) x 100% = 72.1 %
5(d)	Test: Add Br_2 (aq) Observation: Reddish brown Br_2 decolourised and white ppt observed.

6(a)		Type of hybridisation: sp^2  <u>Bonding in an C=C</u>
6(b)	**(i)**	Electrophilic addition 
	(ii)	
6(c)		 The slow step will be the same and the same carbocation will be generated. Since water is present in large quantity, it will act as a competing nucleophile (the oxygen in the water serve as the nucleophile) to attack the carbocation in the fast step forming the product drawn above.
6(d)	**(i)**	 Step 1: Cold, dilute alkaline $KMnO_4$ Step 2: $KMnO_4$ in dilute H_2SO_4, heat (reflux)
	(ii)	 Step 1: Dry HBr (g), room temperature Step 2: Excess ethanolic ammonia, heat in sealed tube

Paper 3 Worked Solutions

1a	**(i)**	Na_2O (s) + H_2O (l) → 2NaOH (aq) pH = 13
	(ii)	P_4O_6 (s) + $6H_2O$ (l) → $4H_3PO_3$ (aq) pH = 4 /3
1b		Amt of $S_2O_3^{2-}$ = $\frac{16.7}{1000} X\, 0.2$ = 3.34 x 10^{-3} mol $Na_2O_2 \equiv H_2O_2 \equiv I_2 \equiv 2S_2O_3^{2-}$ Amt of Na_2O_2 = ½ X 3.34 x 10^{-3} = 1.67 x 10^{-3} mol Mass of Na_2O_2 = 1.67 x 10^{-3} X (23 +23 +32) = 0.1303 g Mass of Na_2O = 0.5 – 0.1303 = 0.3697 g Amt of Na_2O = $\frac{0.3697}{(23+23+16)}$ = 5.964 x 10^{-3} mol Ratio = $\frac{moles\ of\ Na_2O}{moles\ of\ Na_2O_2}$ = $\frac{5.964x10^{-3}}{1.67x10^{-3}}$ = 3.57
1c		H – O – O – H ⇌ H^+ + HO_2^- H_2O ⇌ H^+ + OH^- Presence of extra oxygen, which is electronegative, allows dispersion of negative charge ⇨ Increases stability of conjugate base ⇨ Stronger acid for H_2O_2 CH_3COOOH ⇌ H^+ + CH_3COOO^- CH_3COOH ⇌ H^+ + CH_3COO^- No delocalisation of negative charge over 3 oxygen atoms ⇨ decreases stability of conjugate base ⇨ weaker acid for CH_3CO_3H
1d	**(i)**	BE(C-C) + 2BE(C-O) = 350 + (2 X 360) = 1070 kJ mol^{-1}
	(ii)	ΔH = ∑bonds broken in reactants - ∑bonds formed in products -354 = [BE(C=C) + 4BE(C-H)] – [BE(C-C) + 4BE(C-H) + 2BE(C-O)] -354 = 610 – (BE(C-C) + 2BE(C-O)) BE(C-C) + 2BE(C-O) = 964 kJ mol^{-1} The magnitude of the theoretical values is <u>bigger</u> than that of the actual values as the carbon atoms in the 3-member ring experiences **<u>angle strain</u>**. Since the **<u>bond angles are 60° instead of 109.5°</u>**. This **<u>weakens the C-C and C-O bonds</u>**, **<u>causing the magnitude of its</u>** **<u>bond energy to decrease</u>**, as the ring will break to relieve the angle strain.
1e	**(i)**	C:  E:  F:  (E and F are interchangeable)

	(ii)	Cl_2 (g), uv light, rtp	**(iii)**	 G:	**(iv)**	 D:
	(v)	Reaction II: alcoholic KCN, heat Reaction III: HCl (aq), heat	**(vi)**	 A:		

2a	$\Delta H_{latt} = \Delta H_f(LiH) - [\Delta H_a(Li) + \Delta H_a(H) + 1^{st}\ IE(Li) + 1^{st}\ EA(H)]$ $= (-90.5) - [159.5 + (436 / 2) + 519 + (-73)]$ $= -914\ kJ\ mol^{-1}$		
2b	$AlCl_3\ (s) + 4LiH(s) \rightarrow LiAlH_4\ (s) + 3LiCl\ (s)$ $\Delta H_{rxn} = \sum\Delta H_f(product) - \sum\Delta H_f(reactant)$ $-276 = [(3 \times -408.5) + \Delta H_f(LiAlH_4)] - [(4 \times -90.5) - 704]$ $\Delta H_f(LiAlH_4) = -116.5\ kJ\ mol^{-1}$		
2c	**(i)**	$\Delta G = \Delta H - T\Delta S$ $-27.68 = 3.46 - (298 \times \Delta S)$ $\Delta S = 0.104\ kJmol^{-1}K^{-1}$ ΔS is positive, there is an increase in disorderliness as there is an increase in number of gaseous molecules from 0 to 3.	
	(ii)	oxidation number of H in $LiAlH_4$: -1 Oxidation number of Al in Li_3AlH_6: +3	**(iii)** complex anion: $[AlH_6]^{3-}$ shape: octahedral
2d	**(i)**	 H:  J:  K:	
	(ii)	 L:  M:	
	(iii)	There is no chiral carbon present in M and a plane of symmetry is present.	
2e	**(i)**	$LiAlH_4 + 4H_2O \rightarrow Li(OH) + Al(OH)_3 + 4H_2$ $H^+ + OH^- \rightarrow H_2O$	
	(ii)	Amt of $LiAlH_4$ = $\dfrac{2}{(6.9+27+4)}$ = 0.05277 mol Amt of H_2 = 4 X 0.05277 = 0.2111 mol Vol of H_2 = 0.2111 X 24 = 5.07 dm^3	

3a

Concentrated H_3PO_4 is a weaker oxidising agent than concentrated H_2SO_4.

Concentrated H_2SO_4 is able to oxidise HI to I_2, but not concentrated H_3PO_4

For NaCl: $NaCl(s) + H_2SO_4(l) \rightarrow HCl(g) + NaHSO_4(s)$

For NaI, in the presence of concentrated H_2SO_4:

$NaI(s) + H_2SO_4(l) \rightarrow HI(g) + NaHSO_4(s)$
$2HI(g) + H_2SO_4(l) \rightarrow I_2(g) + 2H_2O(l) + SO_2(g)$
$6HI(g) + H_2SO_4(l) \rightarrow 3I_2(g) + 4H_2O(l) + S(s)$
$8HI(g) + H_2SO_4(l) \rightarrow 4I_2(g) + 4H_2O(l) + H_2S(g)$

For NaI, in the presence of concentrated H_3PO_4:
$NaI(s) + H_3PO_4(l) \rightarrow HI(g) + NaH_2PO_4(s)$

3b (i) Solid P and limited gaseous Cl_2 are heated to form PCl_3 (l) under reflux. PCl_3 is continuously removed by distillation to prevent formation of PCl_5.

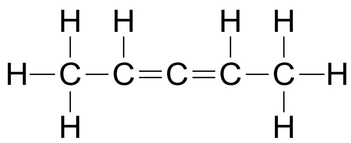
(ii) $PCl_3 (l) + 3H_2O (l) \rightarrow 3HCl (g) + H_3PO_3 (aq)$

PCl_3 undergoes hydrolysis due to the presence of vacant d orbitals, producing white HCl fumes.

3c (i) $POCl_3$

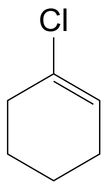
(ii) Q: $CH_3CH{=}C{=}CHCH_3$

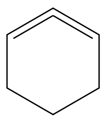
(iii)



bond angles of carbon atoms from left to right: 120, 180, 120

(iv)

R: 

 is unstable (cannot be formed) due to the presence of 2 double bonds side by side within a cyclic ring which results in ring strain.

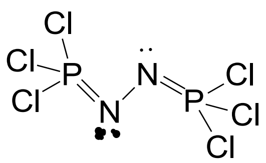
Or

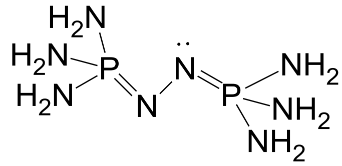
- The **<u>p-orbital of the chlorine atom interacts with the π electron cloud of the C=C double bond in the ring</u>**.
- This **<u>strengthens the carbon-chlorine (C–Cl) bond in the alkene, giving partial double bond character</u>**.

3d (i)

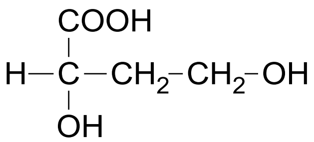
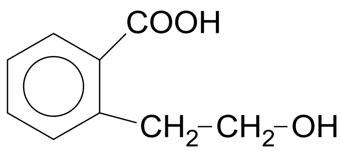
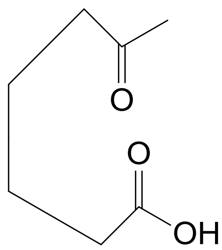
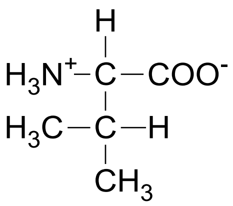
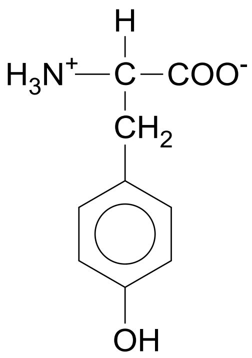
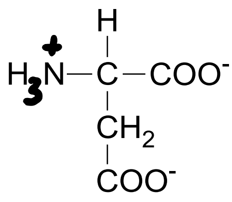
	P	N	Cl
%	20.5	9.2	70.3
Ar	31	14	35.5
Amt (in mol)	0.661	0.657	1.98
ratio	1	1	3

Empirical formula: $PNCl_3$
Molecular formula: $P_2N_2Cl_6$

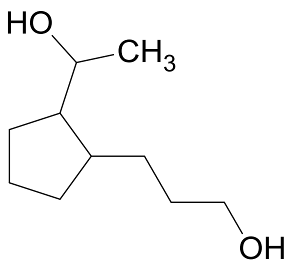


	(ii)	$P_2N_2Cl_6 + 12NH_3 \rightarrow P_2H_{12}N_8 + 6NH_4Cl$	(iii)	T: 

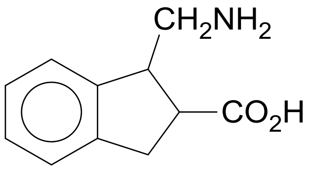
4a	(i)	Electron withdrawing Cl group disperses the negative charge on the conjugate base, stabilising the conjugate base and hence, chloroethanoic acid is a stronger acid with a lower pKa value.
	(ii)	For ethanoic acid, $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ $K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$ $\frac{[CH_3COO^-]}{[CH_3COOH]} = \frac{K_a}{[H^+]} = \frac{10^{-4.76}}{10^{-3.8}} = 0.110$ For chloroethanoic acid, $ClCH_2COOH \rightleftharpoons ClCH_2COO^- + H^+$ $K_a = \frac{[ClCH_2COO^-][H^+]}{[ClCH_2COOH]}$ $\frac{[ClCH_2COO^-]}{[ClCH_2COOH]} = \frac{K_a}{[H^+]} = \frac{10^{-2.87}}{10^{-3.8}} = 8.51$

4b	(i)	U:  Step I: aq HCl, heat Step II: $K_2Cr_2O_7$ / $KMnO_4$, H_2SO_4, heat	(ii)	V:  Step I: aq HCl, heat Step II: $KMnO_4$, H_2SO_4, heat	(iii)	W:  Step I: $KMnO_4$, H_2SO_4, heat Step II: I_2 in aq NaOH, warm, followed by aq HCl, heat
4c	(i)		(ii)		(iii)	

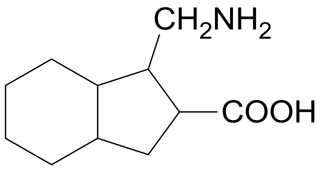
4d	(i)	aspartic acid, ionic interactions	(ii)	valine, van der Waals forces of attraction

Q5a	**(i)**	Cu^{2+}: [Ar] $3d^9$ **(ii)** Ni: [Ar] $3d^8\ 4s^2$
5b		$NH_3\ (aq) + H_2O\ (l) \rightleftharpoons NH_4^+\ (aq) + OH^-\ (aq)$ $Cu^{2+}\ (aq) + 2OH^-\ (aq) \rightleftharpoons Cu(OH)_2\ (s)$ Eqn (1) ○ When dilute NH_3 is added gradually, $[OH^-]$ increases. ○ Ionic product of $Cu(OH)_2$ > K_{sp} of $Cu(OH)_2$ ○ Pale blue precipitate of $Cu(OH)_2$ is formed. $[Cu(H_2O)_6]^{2+}\ (aq) + 4NH_3\ (aq) \rightleftharpoons [Cu(NH_3)_4(H_2O)_2]^{2+}\ (aq) + 4H_2O\ (l)$ Eqn (2) deep blue ○ When excess NH_3 is added, **NH_3** ligands **replace H_2O** ligands to form the **more stable deep blue $[Cu(NH_3)_4(H_2O)_2]^{2+}$** complex. ○ **$[Cu^{2+}]$ decreases** as it is used to form the complex. ○ The equilibrium position in Eqn (1) shifts to the **left** to **increase $[Cu^{2+}]$**. ○ Thus, the pale blue precipitate dissolves.
Q5c	**(i)**	E^0cell = 0.34 – (-0.25) = +0.59 V
	(ii)	$Ni^{2+} + 2e \rightleftharpoons Ni$ E^0oxid When $[Ni^{2+}]$ decreases, by LCP, the position of the above equilibrium will shift left to increase $[Ni^{2+}]$. E^0oxidation will be more negative. E^0cell will be more positive.
Q5d		$E^0(Ni^{2+}/Ni)$ is more negative than $E^0(Cu^{2+}/Cu)$. Ni will be oxidised together with Cu to form Ni^{2+} ions in electrolyte. $E^0(Ag^+/Ag)$ is more positive than $E^0(Cu^{2+}/Cu)$. Ag will not be oxidised. It will drop to the bottom as anode mud. At cathode, only Cu^{2+} is reduced due to more positive E^0 value.
Q5e	**(i)**	Q = It = 2 x23 x 60 = 2760 C Amount of Cu deposited at cathode = 2760 /2x 96500 = 0.0143 mol Expected increase in mass of Cu at cathode = 0.0143 x 63.5 = 0.908 g(3 s.f.)
	(ii)	Mass of Ag removed from anode = 0.0500 g Amount of $Ni(C_4H_7N_2O_2)_2$ = 0.0492 /288.7 = 1.70×10^{-3} mol $Ni(C_4H_7N_2O_2)_2 \equiv Ni^{2+} \equiv Ni$ ∴ Actual mass of Ni removed from anode = (1.70×10^{-3}) x 58.7 = 0.100 g Hence, actual mass of Cu removed = 0.950 -0.100-0.0500 = **0.800 g**
Q5f		X :  Y: $CH_3CH_2CH_2COOH$

Z:



Ω:



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